

Phase	Field	Field key	Data type	Shows/enabled when	Parent fields	Condition (region)	Enabled	Required	Options from	Notes/Warnings
System Information	BAS Controller Manuf.	bas_ctrl_manuf	combo	always	refrigerant = NOT R744/Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{R744 Glycol Carbon Dioxide Ammonia} +	Yes	Yes	
System Information	Cond. Design TD Mid-Point (°F)	cond_design_td_midpoint	float	refrigerant = NOT R744/Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{R744 Glycol Carbon Dioxide Ammonia} +	Yes			
System Information	Cond. Design TD Mid-Point (°F)	cond_design_td_midpoint	float	refrigerant ~ ~[?]{R744 Glycol Carbon Dioxide Ammonia}	refrigerant	~[?]{R744 Glycol Carbon Dioxide Ammonia} +	No			
System Information	Condenser Elec.	cond_elec	text	refrigerant = NOT Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{Glycol Carbon Dioxide Ammonia} +	Yes			
System Information	Condenser Elec.	cond_elec	text	refrigerant ~ ~[?]{Glycol Carbon Dioxide Ammonia}	refrigerant	~[?]{Glycol Carbon Dioxide Ammonia} +	No			
System Information	Condenser Type	condenser_type	combo	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	No			
System Information	Condenser Type	condenser_type	combo	always			Yes	Yes	Air,Water,Fluid,Adiabatic,Evap,Cond,Evap,Fluid	cannot make a selection in this field. Also, if R744 is selected, the label of this field changes to "Gas Cooler Type" which is correct but then I'm not seeing "Gas Cooler Type" as a field
Project Information	Consultant	consultant	text	always			Yes			Moved to Project information phase (was System Information).
System Information	Defrost Elec.	defrost_elec	text	always			Yes			
System Information	Design DB (°F)	design_db	float	always			Yes	Yes		
System Information	Design Interior DB (°F)	design_interior_db	float	always			Yes			
System Information	Design Interior RH (%)	design_interior_rh	float	always			Yes			
System Information	Design WB (°F)	design_wb	float	always			Yes	Yes		
System Information	Est. Summer THR (MBH)	est_summer_thr	float	refrigerant = NOT Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	Yes			
System Information	Est. Summer THR (MBH)	est_summer_thr	float	refrigerant = NOT Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	No			
System Information	Est. Summer THR (MBH)	est_summer_thr	float	refrigerant ~ ~[?]{Glycol Carbon Dioxide Ammonia}	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	No			
System Information	Est. Total Refrig. Charge	est_total_refrig_charge	float	refrigerant ~ ~+	refrigerant	~+	Referen ce		refcharge_calc,est_total_refrig_charge	All charge values are referenced from the Charge Calc phase - shown here, never entered here.
System Information	Est. Winter THR (MBH)	est_winter_thr	float	refrigerant = NOT Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	Yes			
System Information	Est. Winter THR (MBH)	est_winter_thr	float	refrigerant = NOT Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	No			
System Information	Est. Winter THR (MBH)	est_winter_thr	float	refrigerant ~ ~[?]{Glycol Carbon Dioxide Ammonia}	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	No			
Project Information	50 yr. Extreme DB	extreme_db_50yr	float	refrigerant = Carbon Dioxide	refrigerant	*Carbon Dioxide\$	Yes			Moved to Project information phase (was System Information).
Project Information	50 yr. Extreme DB	extreme_db_50yr	float	refrigerant = NOT Carbon Dioxide+	refrigerant	~[?]{Carbon Dioxide\$} +	No			
Project Information	5 yr. Extreme DB	extreme_db_5yr	float	refrigerant = NOT R744/Carbon Dioxide/Ammonia+	refrigerant	~[?]{R744 Carbon Dioxide Ammonia} +	Yes			Moved to Project information phase (was System Information).
Project Information	5 yr. Extreme DB	extreme_db_5yr	float	refrigerant ~ ~[?]{R744 Carbon Dioxide Ammonia}	refrigerant	~[?]{R744 Carbon Dioxide Ammonia} +	No			
System Information	Fixture & Field Piping (lbs)	fixture_field_piping_lbs	float	refrigerant ~ ~+	refrigerant	~+	Referen ce		refcharge_calc,fixture_field_piping_lbs	
System Information	Flash Tank Sat. Temp (°F)	flash_tank_sat_temp	float	refrigerant = R744	refrigerant	*R744\$	Yes			
System Information	Flash Tank Sat. Temp (°F)	flash_tank_sat_temp	float	refrigerant = NOT R744+	refrigerant	~[?]{R744\$} +	No			
System Information	Flash Tank Sat. Temp (°F)	flash_tank_sat_temp	float	refrigerant = NOT R744+	refrigerant	~[?]{R744\$} +	No			
System Information	Gas Cooler Leaving Refrig Temp (°F)	gas_cooler_leaving_temp	float	refrigerant = R744	refrigerant	*R744\$	Yes			
System Information	Gas Cooler Leaving Refrig Temp (°F)	gas_cooler_leaving_temp	float	refrigerant = NOT R744+	refrigerant	~[?]{R744\$} +	No			
System Information	Gas Cooler Leaving Refrig Temp (°F)	gas_cooler_leaving_temp	float	refrigerant = NOT R744+	refrigerant	~[?]{R744\$} +	No			
System Information	Heat Reclaim Media	heat_reclaim_media	text	refrigerant = NOT Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	Yes			
System Information	Heat Reclaim Media	heat_reclaim_media	text	refrigerant = NOT Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	No			
System Information	Heat Reclaim Media	heat_reclaim_media	text	refrigerant ~ ~[?]{Glycol Carbon Dioxide Ammonia}	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	No			
System Information	Heat Reclaim Use	heat_reclaim_use	combo	refrigerant = NOT Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	Yes		Air/Hot Water,Air & Hot H2O,Glycol Heating,N/A	why are there three of these? I believe there should be able to be just the one.
System Information	Heat Reclaim Use	heat_reclaim_use	combo	refrigerant = NOT Glycol/Carbon Dioxide/Ammonia+	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	No		Air/Hot Water,Air & Hot H2O,Glycol Heating,N/A	I think can be removed
System Information	Heat Reclaim Use	heat_reclaim_use	combo	refrigerant ~ ~[?]{Glycol Carbon Dioxide Ammonia}	refrigerant	~[?]{?}{Glycol Carbon Dioxide Ammonia} +	No		Air/Hot Water,Air & Hot H2O,Glycol Heating,N/A	I think can be removed
System Information	Max 0.4% Monthly Design DB (°F)	max_db_04pct	float	refrigerant = R744	refrigerant	*R744\$	Yes			
System Information	Max 0.4% Monthly Design DB (°F)	max_db_04pct	float	refrigerant = NOT R744+	refrigerant	~[?]{R744\$} +	No			
Project Information	0.4% Monthly Max WB (°F)	max_wb_04pct	float	refrigerant = NOT R744+	refrigerant	~[?]{R744\$} +	Yes			Moved to Project information phase (was System Information).
Project Information	0.4% Monthly Max WB (°F)	max_wb_04pct	float	refrigerant = R744	refrigerant	*R744\$	No			
System Information	Num. Suction Groups	num_suction_groups	int	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	No			SC02/Glycol are secondary systems (a skid on a parent rack) - the whole DX / suction group side does not apply.
System Information	Num. Suction Groups	num_suction_groups	int	always			Yes	Yes	1,2,3	
System Information	Parasitic Suct. Loss (°F)	parasitic_suction_loss	float	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	No			
System Information	Parasitic Suct. Loss (°F)	parasitic_suction_loss	text	always			Yes			
System Information	Primary Refrig. H2R Useful Superheat (°F)	primary_refrig_h2r_superheat	float	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	Yes			
System Information	Primary Refrig. H2R Useful Superheat (°F)	primary_refrig_h2r_superheat	float	refrigerant = NOT Carbon Dioxide/Glycol,+	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	No			Skid-level design value - entered by the user, NOT referenced from the parent rack.
System Information	Primary Refrig. Liquid Temp (°F)	primary_refrig_liquid_temp	float	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	Referen ce		refparent_rack,agprimary_suction_group_serving_skid,liquid	Read from the parent rack, on the suction group selected in 'Primary Suction Group Serving Skid'.
System Information	Primary Refrig. Return Gas Temp (°F)	primary_refrig_return_gas_t	float	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	No			
System Information	Primary Refrig. Return Gas Temp (°F)	primary_refrig_return_gas_t	float	refrigerant = NOT Carbon Dioxide/Glycol,+	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	Referen ce		refparent_rack,agprimary_suction_group_serving_skid,refr	
System Information	Primary Refrig. Sat. Cond. Temp (°F)	primary_refrig_sat_cond_tem	float	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	Referen ce		refparent_rack,sat_condensat,temp	
System Information	Primary Refrig. Sat. Cond. Temp (°F)	primary_refrig_sat_cond_tem	float	refrigerant = NOT Carbon Dioxide/Glycol,+	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	No			
System Information	Primary Refrig. SST (°F)	primary_refrig_sst	float	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	Referen ce		refparent_rack,agprimary_suction_group_serving_skid,sat	
System Information	Primary Refrig. SST (°F)	primary_refrig_sst	float	refrigerant = NOT Carbon Dioxide/Glycol,+	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	No			
System Information	Primary Suction Group Serving	primary_suction_group_serve	combo	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	Yes		SG1,SG2,SG3	NEW field. A skid is added as a sub-system of a rack, this is where the user picks which suction group on that rack serves it. Drives every Primary Refrig. * reference below.
System Information	Primary Suction Group Serving	primary_suction_group_serve	combo	refrigerant = NOT Carbon Dioxide/Glycol,+	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	No			
System Information	Pump Skid H2R Approach (°F)	pump_skid_h2r_approach	float	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	Referen ce		global:skid_models,approach,f	
System Information	Pump Skid H2R Approach (°F)	pump_skid_h2r_approach	float	refrigerant = NOT Carbon Dioxide/Glycol,+	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	No			
System Information	Pump Skid Maximum Capacity (GPM)	pump_skid_max_capacity_gpm	float	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	Referen ce		global:skid_models,max_flow,gpm	
System Information	Pump Skid Maximum Capacity (GPM)	pump_skid_max_capacity_gpm	float	refrigerant = NOT Carbon Dioxide/Glycol,+	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	No			
System Information	Pump Skid Minimum Capacity (GPM)	pump_skid_min_capacity_g	float	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	Referen ce		global:skid_models,min_flow,gpm	
System Information	Pump Skid Minimum Capacity (GPM)	pump_skid_min_capacity_gpm	float	refrigerant = NOT Carbon Dioxide/Glycol,+	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	No			Looked up from the skid_models table by the selected Pump Skid Model - not user-entered. Pump Skid Model itself stays user-selectable.
System Information	Pump Skid Model	pump_skid_model	combo	refrigerant ~ ~[?]{Carbon Dioxide Glycol }	refrigerant	~[?]{Carbon Dioxide Glycol }	Yes			
System Information	Pump Skid Model	pump_skid_model	combo	refrigerant = NOT Carbon Dioxide/Glycol,+	refrigerant	~[?]{?}{Carbon Dioxide Glycol} +	No			

Circuit Designer	#	total_length	int	circuit_type = NOT Case or Custom AND mode = any AND length_unit = any	circuit_type,mode,length_unit	^(?!?Case Custom)\$.*\$.*\$.*\$.*\$	No	
Circuit Designer	#	total_length	int	circuit_type = Case or Custom AND mode = any AND length_unit = any AND qty_2drs_4rt = any AND qty_3drs_6rt = any AND qty_3drs_6rt = any AND qty_4drs_8rt = any AND qty_5drs_12rt = any	circuit_type,mode,length_unit,qty_2drs_4rt,qty_3drs_6rt,qty_4drs_8rt,qty_5drs_12rt	^(?!?Case Custom)\$.*\$.*\$.*\$.*\$.*\$.*\$.*\$.*\$.*\$	Reference	calc(qty_2drs_4rt*if(length_unit="dr" 2.4) + (qty_3drs_6rt*if(length_unit="dr" 3.6)) + (qty_4drs_8rt*if(length_unit="dr" 4.8)) + (qty_5drs_12rt*if(length_unit="dr" 5.12)) + qty_6rt5 + qty_slot6
Circuit Designer	H	walk_in_height	int	circuit_type = Walk in AND mode in (Manual, Database)	circuit_type,mode	^Walk_in\$.*?Manual Database)\$	Yes	Yes
Circuit Designer	H	walk_in_height	int	circuit_type = NOT Walk in	circuit_type	^(?!Walk_in)\$.*\$	No	
Circuit Designer	L	walk_in_length	int	circuit_type = Walk in AND mode in (Manual, Database)	circuit_type,mode	^Walk_in\$.*?Manual Database)\$	Yes	Yes
Circuit Designer	L	walk_in_length	int	circuit_type = NOT Walk in	circuit_type	^(?!Walk_in)\$.*\$	No	
Circuit Designer	W	walk_in_width	int	circuit_type = Walk in AND mode in (Manual, Database)	circuit_type,mode	^Walk_in\$.*?Manual Database)\$	Yes	Yes
Circuit Designer	W	walk_in_width	int	circuit_type = NOT Walk in	circuit_type	^(?!Walk_in)\$.*\$	No	
Circuit Designer	WM Equip. Tag	with equip. tag	text	always			Yes	

Never user-entered - always the sum of the qty columns, in whichever unit length_unit is set to. Slot sizes for Qty 5 / Qty 6 are TBC.